

MANAV WILLIAM 32
ALISHA KHAN 36

WASHINGTON STATE ELECTRIC VEHICLE LANDSCAPE

A Power BI dashboard project — adoption, market concentration, geography, range and pricing

Objective: explore the historical Washington State EV population dataset and communicate adoption, manufacturer concentration, geography, range and pricing patterns through two dashboard pages

Dataset & Source

- Working file: a Kaggle dataset built on Washington State DOL electric-vehicle population records.
- Fields cover vehicle identity (Make, Model, Model Year), location (County, City, Zip, coordinates), EV type, CAFV eligibility, Electric Range and Base MSRP.
- Expected Price was excluded — its values were inconsistent and unreliable for pricing analysis; it is not presented as an official DOL sales price.
- Scope: the dataset is historical, with model years running through 2022 — findings describe the dataset, not the current 2026 EV market.
- Tool: Microsoft Power BI Desktop — Power Query for preparation, Power BI visuals and DAX for analysis and presentation.

Data Cleaning & Preparation

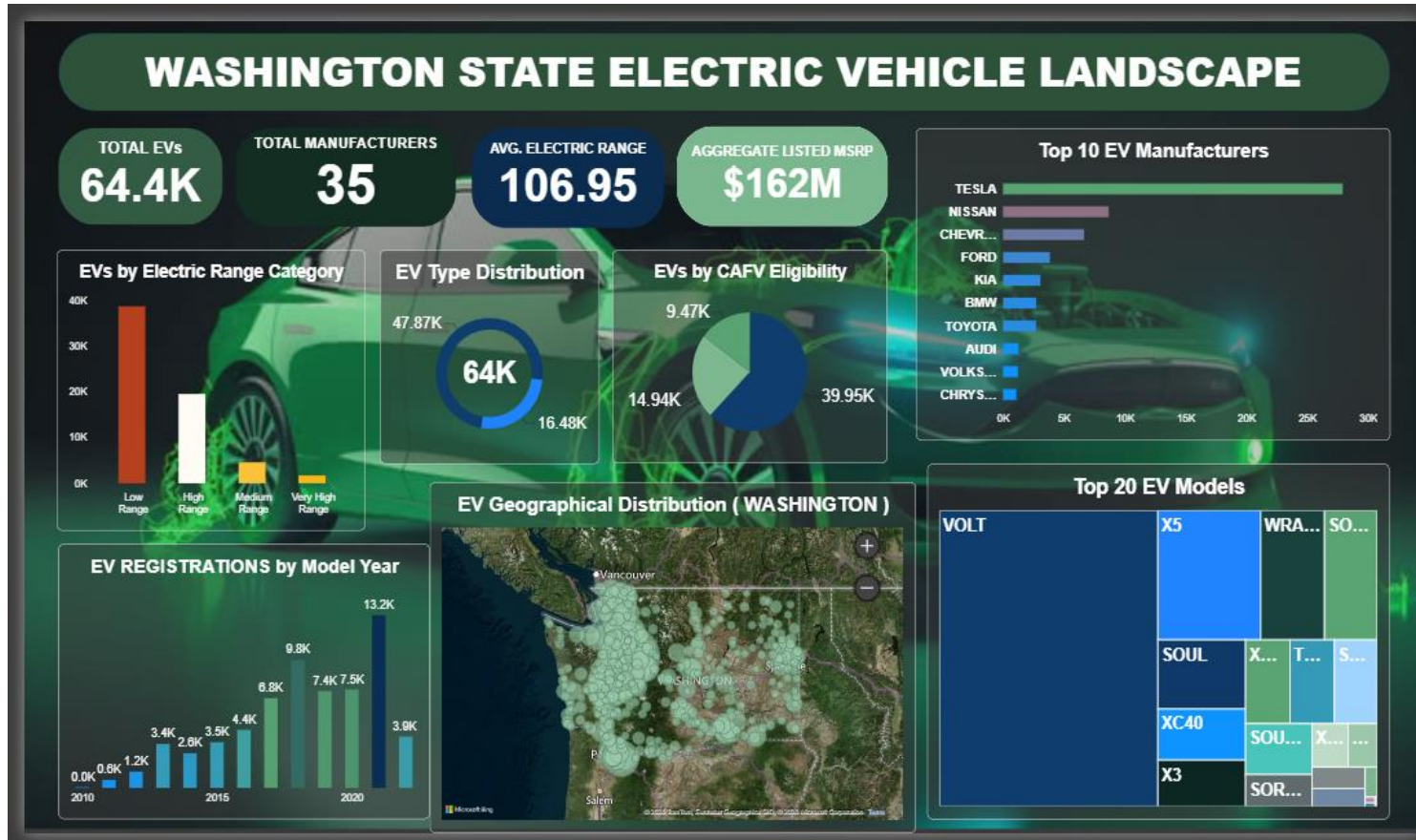
Duplicates	Checked ID, VIN (1-10) and DOL Vehicle ID — none found, no rows removed.
Missing values	Column quality check; 7 blank Model Years handled via a “Year Not Available” group.
Text standardization	Trim/Clean applied to Make, Model, County, City, State, EV Type, CAFV Eligibility, Utility.
Zero values	Electric Range = 0 and Base MSRP = 0 kept, but treated as unavailable/not-reported — not real zeros.
Expected Price	Removed from analysis: values were inconsistent and unsuitable for pricing conclusions.

Feature Engineering

- Model Year Group — buckets years into 2011–2015, 2016–2018, 2019–2020, 2021–2022, plus “Year Not Available” for nulls.
- Range Category — Low / Medium / High / Very High Range, split at 100/200/300 miles.
- MSRP Category — Budget / Mid-Range / Premium / Luxury, with “MSRP Not Available” for zero values.
- MSRP per Range — $\text{Base MSRP} \div \text{Electric Range}$, an exploratory price-per-range metric (null when either input is zero).
- Latitude & Longitude — parsed out of the Vehicle Location POINT string so the map could plot Washington records.

These are project-defined analytical bands, not official industry classifications.

Dashboard Page 1 — The EV Landscape

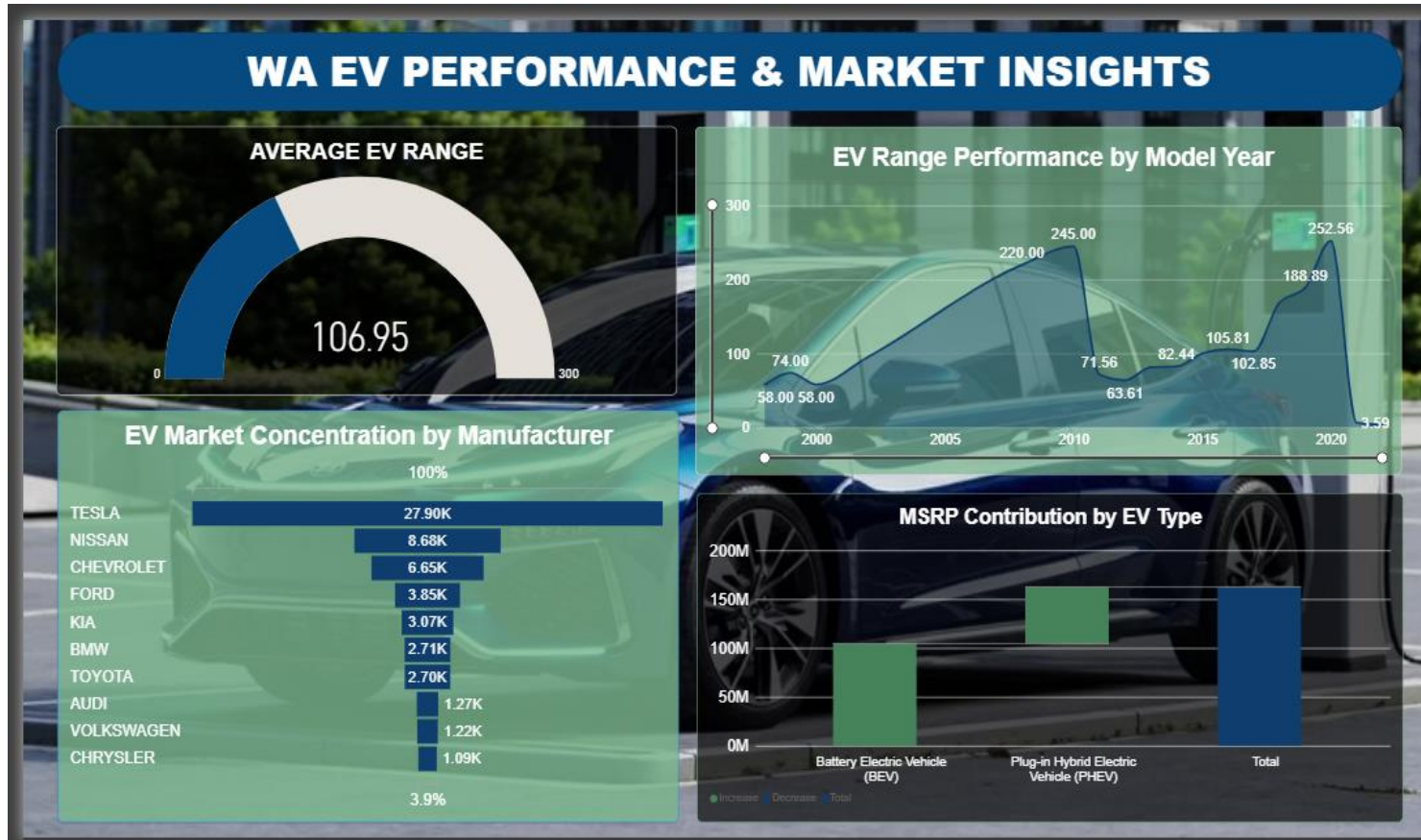


- 4 KPI cards: 64.4K total EVs, 35 manufacturers, 106.95 avg. range, ~\$162M aggregate listed MSRP.
- EV Type Distribution (donut) and CAFV Eligibility (pie) show vehicle-type and eligibility composition.
- Top 10 EV Manufacturers (bar) and Top 20 EV Models (treemap) show concentration among leaders.
- EV Registrations by Model Year (column) tracks record volume over time.
- Geographical map plots EV records across Washington using extracted coordinates.

Page 1 Insights

- Adoption scale: the dataset represents about 64.4K unique EV records across 35 manufacturers.
- Manufacturer concentration: Tesla leads the Top 10 list, followed by Nissan and Chevrolet.
- Model concentration: record volume is concentrated among a smaller set of popular models in the Top 20 treemap.
- Model-year trend: registrations vary by year with a visible peak around 2019.
- Geography: EV records show clear spatial concentration across Washington State.

Dashboard Page 2 — Performance & Market Insights



- Gauge: Average EV Range headline (106.95), scaled 0–350.
- Funnel: EV Market Concentration by Manufacturer — record counts across the Top 10.
- Area chart: EV Range Performance by Model Year — average range across years.
- Waterfall: MSRP Contribution by EV Type — BEV vs. PHEV share of aggregate listed MSRP.

Page 2 Insights

- Manufacturer concentration: Tesla holds ~27.90K records — well ahead of Nissan (~8.68K) and Chevrolet (~6.65K).
- Range by model year is uneven, not steadily rising — higher averages appear around 2010 and again 2019–2021.
- The 2022 point (3.59) is unusually low and likely reflects few records or unavailable range values that year, not a technology regression.
- Both BEV and PHEV contribute to aggregate listed MSRP, with PHEV contributing the larger share in the current view.
- All figures describe listed prices and dataset averages — not sales revenue or real-time market data.

Business & Market Implications

- Manufacturer presence: heavy concentration around one leading manufacturer highlights where market share and brand visibility are strongest in this dataset.
- Vehicle mix: the BEV/PHEV split and range categories help frame product-positioning and infrastructure-planning questions.
- Range characteristics: uneven range-by-year patterns suggest performance claims should be checked against record counts, not read as a smooth trend.
- Listed-price composition: comparing MSRP contribution by EV type supports pricing-tier and positioning discussions, distinct from actual sales revenue.
- Overall, the dashboard supports descriptive, exploratory market understanding — not causal or predictive claims.

Limitations & Conclusion

- Historical scope: data runs through model year 2022 — not a real-time 2026 snapshot.
- Listed MSRP $\neq$ revenue: aggregate MSRP reflects listed prices summed across records, not sales value.
- Top-N visuals (Top 10 manufacturers, Top 20 models) show only the leaders, not the full distribution.
- Year-over-year averages can be sensitive to record counts — avoid causal “technology improved because of year” claims.

Overall, the dashboard gives a structured view of the Washington EV population — combining adoption, manufacturer concentration, geography, vehicle type, model-year performance and listed MSRP — built for descriptive insight and visual storytelling, not causal or real-time forecasting.